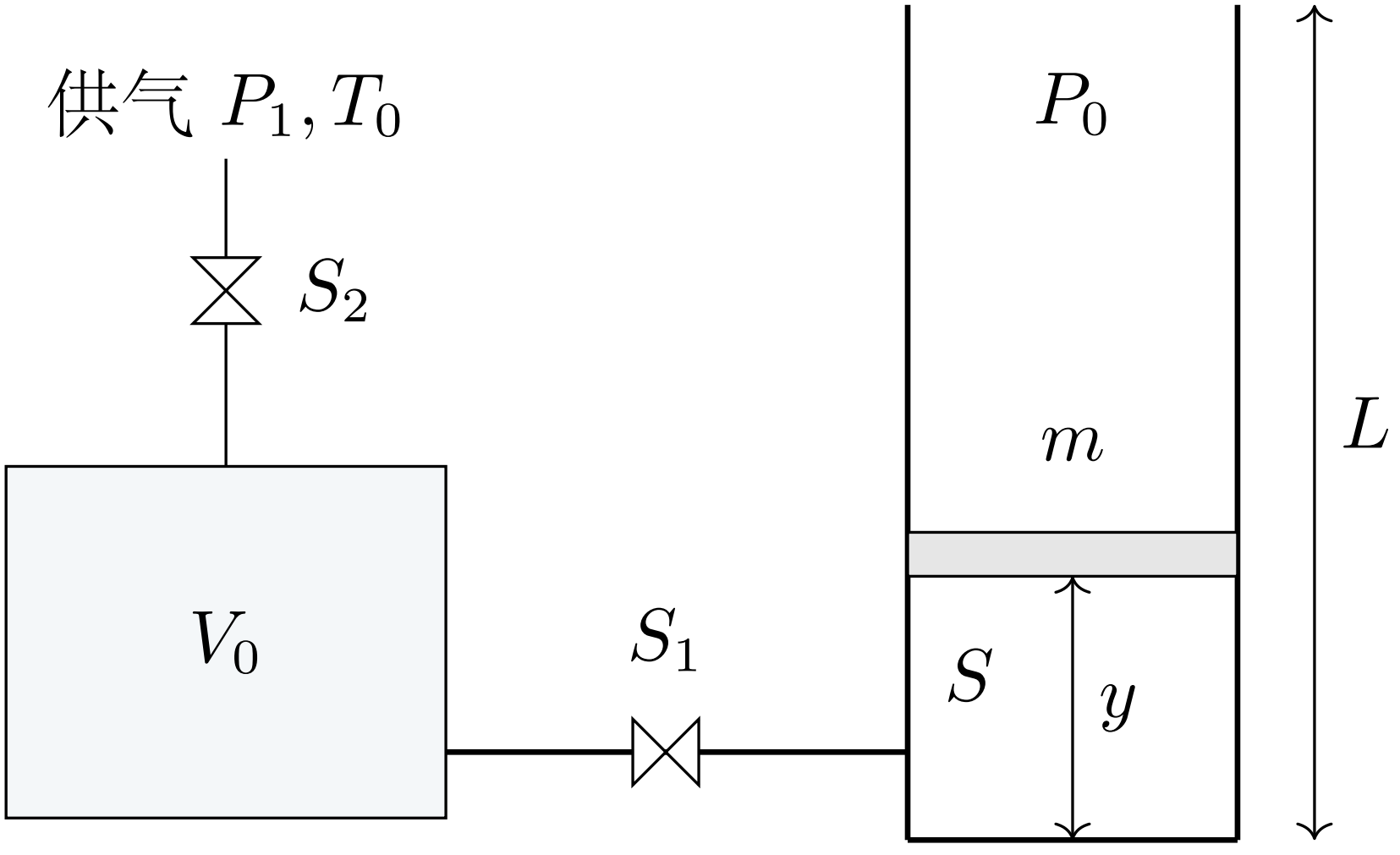
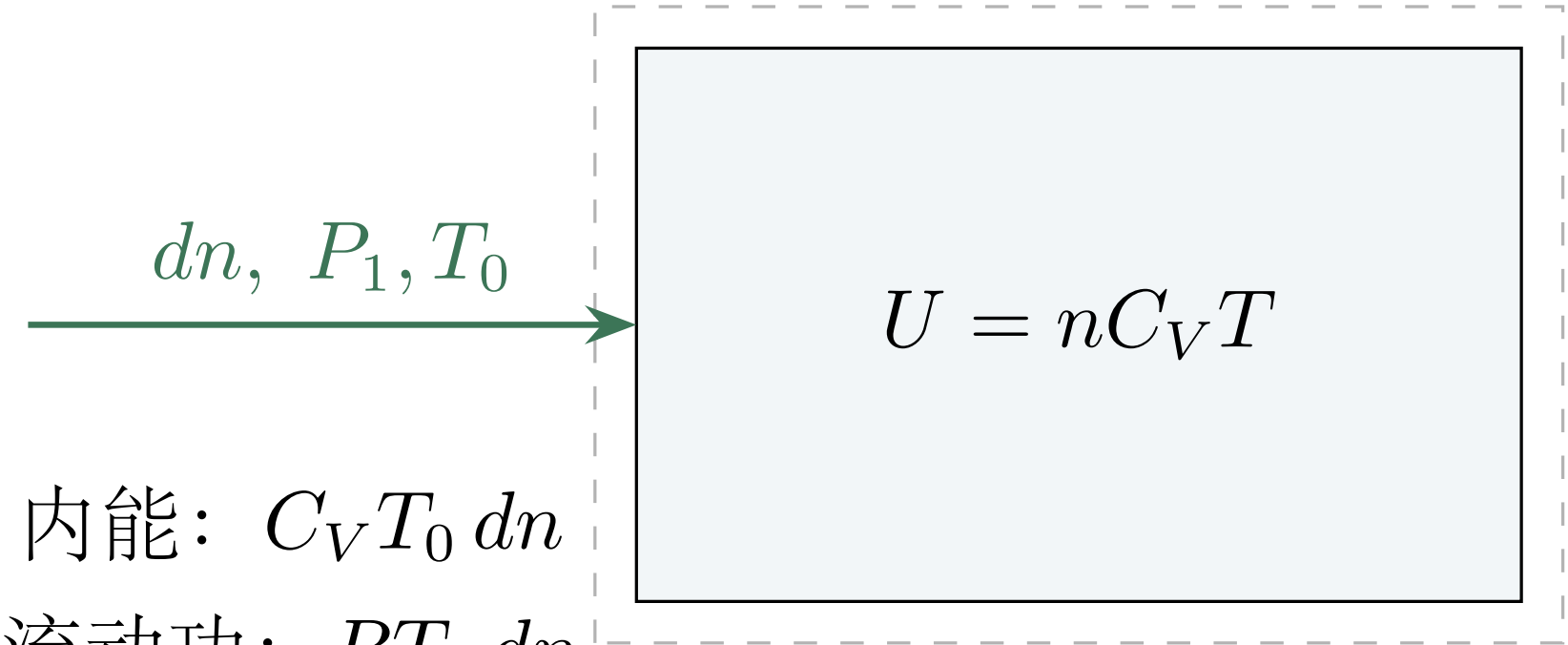


发射阶段：S_2 关闭，S_1 打开

刚性、绝热，初始真空

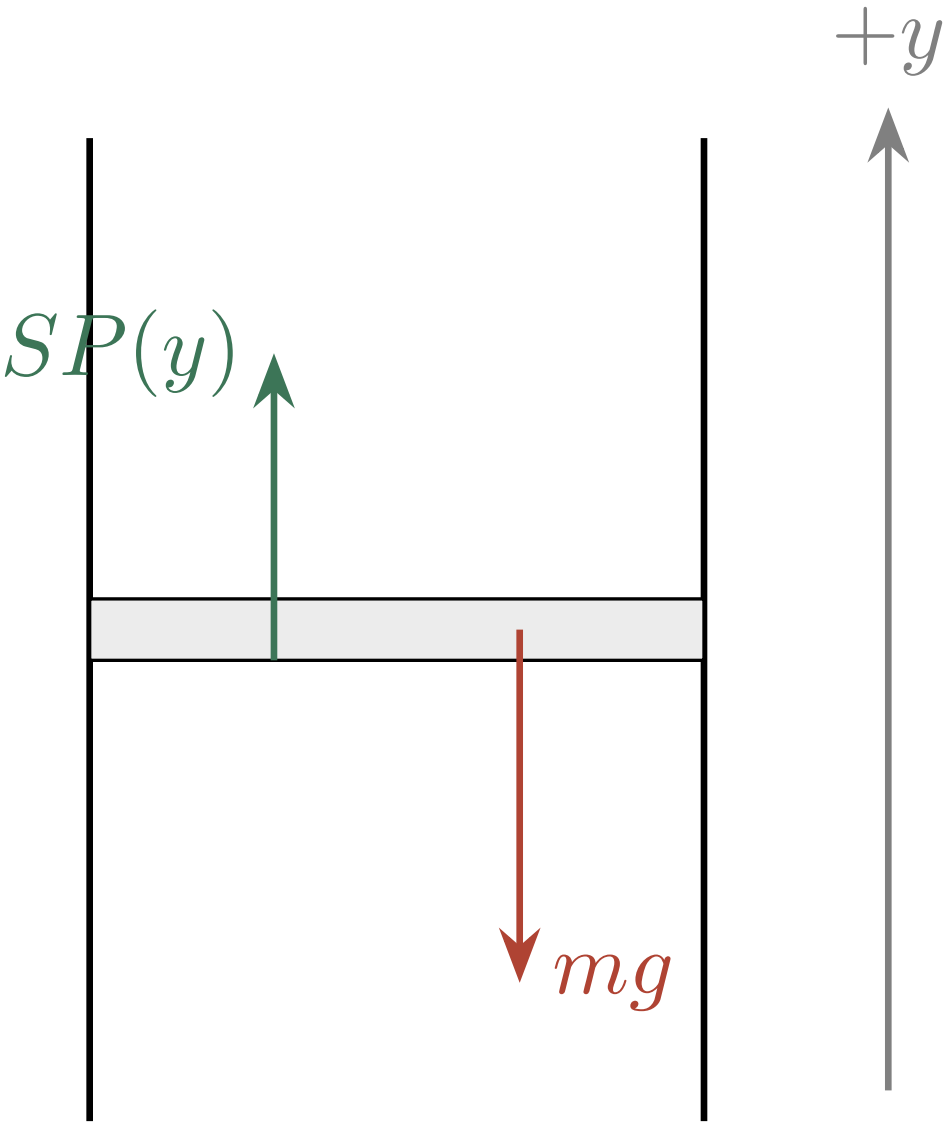


内能：$C_V T_0\, dn$

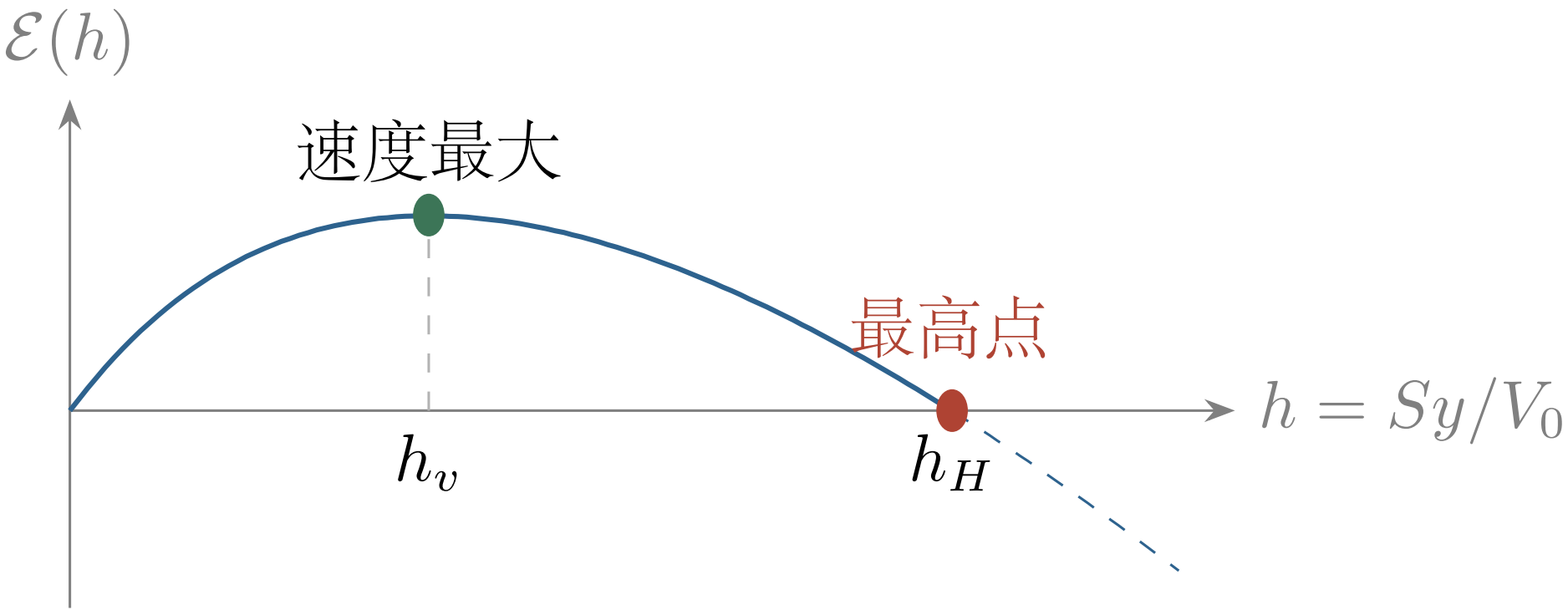
流动功：$RT_0\, dn$

输入总能量 $C_P T_0\, dn$；不是仅输入内能

$$m\ddot{y} = SP(y) - mg$$



下方气体体积 $V_0 + Sy$



示例 $\alpha = 2$; $\mathcal{E} = \frac{1}{2}mv^2/(mgV_0/S)$